

Storage Decision Matrix

Criteria	S3	EBS (gp3)	EBS (io2)	EFS	FSX for Linux
Type	Object	Block	Block	File (NFS)	File (Linux)
Access pattern	HTTP API	Single EC2	Single EC2	Multi- EC2	Multi- EC2 (HPC)
Max size	Unlimited	16 TB	64 TB	Unlimited	Unlimited
Durability	99.9999 99.9999%	99.8 - 99.9%	99.99 9%	99.9999 99.999%	N/A (scratch)
Use when	Static assets, backups, data lakes	Boot vols, databases	High-IOPS databases	Shared app configs, CMS	ML, HPC
Don't use when	Need file system mount	Need shared access	Crit- sensitive	Single instance only	Standard web apps
Base cost	\$0.003/GB -mo	\$0.08/ GB-mo	\$0.125/ GB-mo	\$0.30/ GB-mo	\$0.14/ GB-mo
Cost per TB/yr	\$282.62	\$983.04	\$1,536	\$3,686.40	\$1,720.32

EBS Volume Types

Type	IOPS	Throughput	Cost/GB-mo	Use Case
gp3	3,000-16,000	125-1,000 MB/s	\$0.08	Default for everything
gp2	3 IOPS/GB (burst)	250 MB/s	\$0.10	Legacy - no use for new builds
io2	up to 256,000	4,000 MB/s	\$0.125	Databases needing guaranteed IOPS
st1	500	500 MB/s	\$0.045	Big data, log processing
sc1	250	250 MB/s	\$0.015	Cheapest, cold storage

Decision framework:

Need < 1ms latency for databases?
→ io2 Block Express

Need consistent IOPS for general workloads?
→ gp3

Need cheap storage for infrequent access?
→ sc1

Need throughput for sequential reads?
→ st1

Boot volume?
→ gp3 (always)

Actual benchmarks for gp3:

Sequential write $\rightarrow$ 139 MB/s ★
read $\rightarrow$ 136 MB/s
Random 4K IOPS (read) $\rightarrow$ 7,017 ★
(write) $\rightarrow$ 7,076 ★
Avg latency (ms) $\rightarrow$ 677 ★

Actual benchmarks for EFS:

Sequential write $\rightarrow$ 109 MB/s
read $\rightarrow$ 302 MB/s ★
Random 4K IOPS (read) $\rightarrow$ 4,704
(write) $\rightarrow$ 4,727
Avg latency (ms) $\rightarrow$ 3,472

- EFS costs $\sim 4\times$ more per GB than gp3 (0.30 vs 0.08)
- EFS is the only option for shared file system across EC2 instances
- EFS scales automatically, no provisioning needed
- EFS has lifecycle policies too (move infrequent files to IA tier at \$0.036/GB)